

Hydrokinetic power generation for rural electricity supply: Case of South Africa

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ABSTRACT

This study investigates the possibility of using and developing hydrokinetic power to supply reliable, affordable and sustainable electricity to rural, remote and isolated loads in rural South Africa where reasonable water resource is available. Simulations are performed using the Hybrid Optimization Model for Electric Renewable (HOMER) and the results are compared to those from other supply options such as standalone Photovoltaic system (PV), wind, diesel generator (DG) and grid extension. Finally the paper points out some major challenges that are facing the development of this technology in South Africa.

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1. Introduction

South Africa is endowed with abundant renewable energy resources that can be used optimally to help facing the challenges of global warming, reduce green house gases emissions resulting from the extensive use of fossil fuel as primary resource of electric energy and to have an energy security through diversification of supply [1]. It is in this context that the South African Government is giving a push to renewable energy and integrates it into the mainstream energy economy. To reach this goal, South African Government is setting a target 10,000 GWh renewable energy contribution to be produced mainly from biomass, wind, solar and small-scale hydropower by 2013 [2].

Hydrokinetic power generation is a category of hydropower energy that extracts kinetic energy from flowing water rather than potential energy from water fall. Hydrokinetic power systems avoid many of the challenges which are coming across with traditional hydropower, such as high civil infrastructure costs, and the need of acceptable water head [3]. They have simple design and can be easily installed and maintained by local population at low cost if installed in remote and rural areas. Another advantage is that hydrokinetic can be easily installed in free-flowing rivers or streams to enhance energy extraction, these make hydrokinetic far more

competitive compared to traditional micro hydropower even though they can extract almost the same amount of energy.

Approximately 6000–8000 potential sites for traditional micro hydropower applications are situated mainly in Eastern Cape and KwaZulu-Natal provinces [4]. Due to the simplicity of the hydrokinetic power design, there are theoretically huge numbers of potential sites as compared to small hydropower generation. The cost of energy extracted from hydrokinetic is lower than the one of small hydropower. Hydrokinetic technology is more economical compared to solar power system; it is thus a better candidate for South African rural electrification programs where water resource is available.

This study investigate the possibility of using and developing hydrokinetic power to extend the reliable, affordable and sustainable electricity supplies for rural and remote loads in South Africa where reasonable water resource is available. For this purpose, we have selected a potential site from which we have acquired data such as water flow and energy demand needed as input to the HOMER program. The simulation results of the proposed hydrokinetic system are compared to those from other power supply options such as standalone PV, wind, diesel generator and grid extension line to find the optimal and most suited option to supply the rural and isolated load.

2. Hydropower situation in South Africa

2.1. Hydropower potential

By international standards, the extensive development of hydropower for electricity generation has not yet been considered

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Table 1
Hydropower potential in South Africa.

Size	Type	Installed capacity (MW)	Estimated potential (MW)
Macro hydropower (larger than 10 MW)	(i) Imported	1450	36,400
	(ii) Pumped storage for peak supply	1580	10,400
	(iii) Diversion fed	—	5200
	(iv) Dam storage regulated head	662	1520
	(v) Run of river	—	270
Small hydropower (from a few kW to 10 MW)	As above (iv) and (v)	29.4	113
	Water transfer	0.6	38
	Refurbishment of existing plants	8.0	16
	Gravity water carrier	0.3	80
Sub-total for all types		3730.3	53,837
Excluding imported from abroad		2280.3	17,437
Excluding pump storages using coal based energy		700.3	7237
Total “green” hydro energy potential available within the border of South Africa			7237

seriously in South Africa. No significant development of hydropower in the country has been noted for 30 years, except the new small-scale installation of 7 MW capacity commissioned at the Sol Plaatjie Municipality Free State province. At the present the overall hydro-electricity generation capacity represents only about 5% of present total 45,500 MW installed generation capacity [5]. Table 1 below gives a summary of the hydropower potential in South Africa.

2.2. Where to look for hydroelectricity in South Africa

The rural communities in the Eastern Cape, Mpumalanga and KwaZulu-Natal provinces have access to water resources with good hydropower potential [6]. The development of small-scale traditional hydroelectric installation particularly for the commercial and domestic consumption should be strongly promoted and supported. Communities with hydropower potential and interest in developing hydroelectricity needs a wide professional support since any new hydropower installation is costly and requires technical and operational inputs from civil, mechanical and electrical professionals. The figure below shows the areas where potential site for development of micro-hydropower as well as the location where they have already been implemented [7].

We have to notice that Fig. 1 and Table 1 do not take into consideration the energy potentially available from hydrokinetic which can represent a potential source of electric power even greater than the one from micro- and pico-hydropower plants. The ideal location for a hydrokinetic turbine is to be located in deep strong flowing rivers or immediately downstream from an existing conventional hydropower plant where electric transmission wires and interconnection facilities are located, and also where the energy remaining in the water current existing from the turbines in the dam can be reused. Theoretically, a greater number of potential sites to implement hydrokinetic power can be identified compare the traditional small-scale hydropower.

3. Hydrokinetic power

3.1. Technology

Hydrokinetic was originally developed to surmount the numberless of problems associated with dams throughout the world. This system is erected into the river or stream which results in the following advantages compared to the traditional hydropower:

- No dam,
- No destruction of nearby land,
- No change in the river flow direction,
- Reduction of flora and fauna destruction.

The terms hydrokinetic encapsulate both tidal and river applications. Within the context of this paper, the focus is on river application, since it is suitable for energy generation at remote and isolated locations.

3.2. The turbine

Most of the operation principles of the hydrokinetic turbines are based upon wind turbines, as they work in a similar way but with the possibility of having close to 1000 time more energy from the hydrokinetic compared to the wind turbine of the same swept area [8]. The power available (P_a) in watts can be worked out using the following equation.

$$P_a = \frac{1}{2} \times A \times \rho \times V^3 \times C_p \quad (1)$$

A = area in metres squared (m^2)
 ρ = density of water (1000 kg/m^3)
 V = velocity of water (m/s)
 C_p = the power coefficient

The theoretical maximum power available from the river is expressed by the equation above using a power coefficient of 0.592 or 59% efficiency. But a small-scale river turbine has its own losses which will reduce the power coefficient to around 0.25.

From equation (1) above, it is noticeable that the power increases in a cubed relationship to the velocity of the flow of water

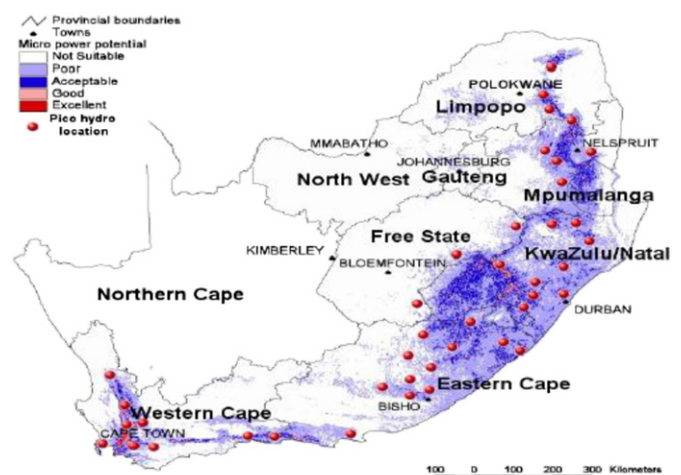


Fig. 1. Small-scale hydropower distribution in South Africa.

Table 2
Domestic power demand estimation.

Equipment	Amount	Power (kW)	Time (h)	Energy (kWh/d)
Light	5	0.006	6	0.18
Radio	1	0.020	5	0.1
T.V.	1	0.07	5	0.35
Iron	1	1	0.1	0.1
Kettle	1	1.5	0.05	0.075
Fridge	1	0.12	24	2.88
Phone charger	3	0.004	1	0.012

past the turbine. Therefore it is important to find the best flow to get the best power output.

3.3. Generator

In order to reduce costs, and to be able to rely on locally-made technology, permanent magnet generator can be used. The magnets allowed the speed of generation to be reduced, and lowered the cost of the equipment, which itself could be adapted to be a river turbine rotor and ultimately, tested and built [9]. Due to lower generation speed, gearboxes or generators with high number of poles can be used [10].

4. System design

The HOMER simulation program has been chosen as a tool for system design. HOMER was selected due to its capability to evaluate the best option by harnessing energy from a single or combination of various energy resources [11]. It is an economic model that provides rational selection of the most cost effective option [12]. Furthermore, its hourly energy flow approach offers a comprehensive analysis of the system performance throughout a year. Two case studies have been conducted on different sites from which the load energy demand, the renewable energy resources, as well as the cost of the supply options (hydrokinetic, solar PV, wind, diesel generator and grid extension) have been used as input to HOMER.

4.1. Case 1: rural household

4.1.1. Load description

For this first case, a typical rural household in the KwaZulu-Natal has been selected.

The site is situated at 30.6 Latitude South and 29.4 Longitude East.

Table 2 gives domestic appliances, power demand and running times for an average typical household in rural South Africa [13]. The load is 3.4 kW peak and 9.5 kWh per day.

Table 3
Site 1 energy resources.

Month	Water speed (m/s)	Daily radiation (kWh/m ² /d)	Wind speed (m/s)
January	5.31	6.23	4.1
February	7.25	5.83	3.9
March	6.09	5.21	3.8
April	1.81	4.46	3.9
May	2.67	3.81	4.1
June	2.18	3.33	4.5
July	1.84	3.62	4.5
August	1.54	4.29	4.6
September	1.41	5.08	4.8
October	1.69	5.41	4.6
November	2.83	6.00	4.3
December	5.27	6.35	4.0
Average	3.32	4.947	4.26

Table 4
BTS load.

Items	Power consumption (kW)	Usage h/day
Constant site load (BTS, TX)	2	24
Air-conditioner (12,000 BTU)	1.8	6
Air-conditioner start-up	3.3	–

4.1.2. Resources assessment

The summary of the water velocity [14], wind velocity and solar radiation [15] from the site is shown in Table 3.

The theoretical potential power available from the hydrokinetic turbine (P_a) can be found with the help of equation (1) using the following characteristics from the selected stream:

- Minimum water velocity in the worst month: 1.41 m/s
- Viable depth: 1.8 m
- Width: 5.2 m
- Cross sectional area: 9.36 m²
- $P_a = 1075$ kW

A correction factor of 0.8 has been applied to the measured values to accommodate friction effects along the bottom and sides of the river on the current velocity [16]. With reasonable sizing of the battery storage system, this available power can cover the load energy requirement without interruption. The selected site has very good solar and wind resources as shown in Table 3, so the solar PV system, wind and the standalone diesel generator can be compared to the hydrokinetic while supplying the same load to find out which one is the best supply option for the site.

4.2. Case 2: base transceiver station

4.2.1. Load description

The medium-sized indoor base transceiver station used has an equipment power loading of 2 kW. The items and their power consumptions are given in Table 4. Normally the full load will only be the constant site load and the air-conditioner running power (or 3.8 kW), or when temperatures permit the air-conditioners to be shut off, only the BTS load (2 kW) [17]. The total power required in the worst case will be the full load plus the air-conditioner start-up power (7.1 kW). Thus the load is 7.1 kW peak and 58.8 kWh energy consumption per day.

4.2.2. Resources assessment

The summary of the water velocity [14], wind velocity and solar radiation [15] from the site is shown in Table 3 plus the air-conditioner start-up power (7.1 kW). Thus the load is 7.1 kW peak and 58.8 kWh energy consumption per day.

Table 5
Site 2 energy resources.

Month	Water speed (m/s)	Daily radiation (kWh/m ² /d)	Wind speed (m/s)
January	6.410	8.44	6.6
February	5.270	7.50	5.9
March	3.830	6.22	5.8
April	3.120	4.66	5.1
May	2.470	3.43	4.9
June	2.160	3.01	5.3
July	1.580	3.21	5.1
August	1.220	4.10	5.3
September	1.710	5.33	5.6
October	2.430	6.82	6.2
November	4.190	7.96	6.2
December	6.600	8.51	6.0
Average	3.047	5.76	5.7

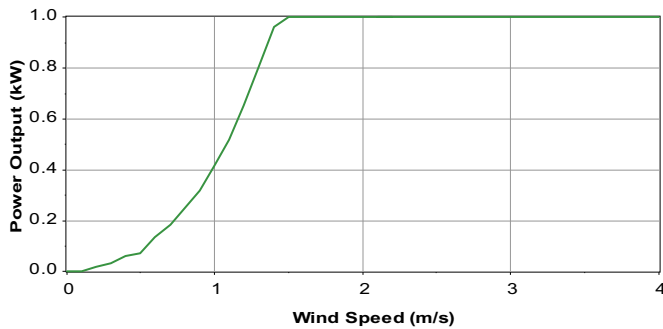


Fig. 2. Hydrokinetic power-curve.

The site is located near Cape Columbine at 32.8 Latitude South and 17.9 Longitude East with one of the best solar potential in South Africa and very good yearly wind speed. The water potential is in favor of the development of hydrokinetic system. The renewable resources can be found on Table 5.

4.3. Components information

4.3.1. Hydrokinetic power

Unfortunately, HOMER is not equipped with a hydrokinetic power module as considered in this study. Consequently, instead of using a traditional micro-hydro module, the wind turbine component has been used with related hydrokinetic input rather than wind-related information [14]. This approach was considered because wind turbines share some similarities with hydrokinetic turbines which are commonly referred to as 'underwater wind turbines'. Thus, the wind turbine power-curve has been replaced with the power-curve of the selected hydrokinetic turbine by altering the wind speed information with the river current velocity [18].

The Darrieus hydrokinetic turbines (DHT) developed by Alternative Hydro Solutions in Canada has been chosen because of its simple structure and its ability to generate a relatively high power output from low to medium flow velocities [19]. Figs. 2 and 3 are the power-curve of the turbine based on information from the turbine's manufacturer. The turbine's rated power is 1 kW at 1.4 m/s current velocity. Since the information about the power outputs at the flow speed above 1.5 m/s flow was not available, it has been assumed that above 1.5 m/s, there are no increases in power output.

The investments as well as the replacement costs of the 1 kW hydrokinetic turbine are \$7500 respectively, the operation and maintenance cost is \$20/year; the system lifetime span is 25 years [18].

4.3.2. Photovoltaic system

The actual price of the PV module is set at 3.59 \$/W in USA, considering the transport and other unpredictable costs. The price of PV is set to 4500 \$/kW with the replacement cost of \$4100. The cost of the inverter is set 800 \$/kW [20]. The operation and maintenance costs of the photovoltaic module and of the inverter are estimated at 105 and 10 \$/yr respectively [21]. The price of the deep cycle battery is \$215 with a replacement cost of \$215, the operation and maintenance is \$5 [22], the lifetime is taken as 20 year.

	G1	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
	2	4	3.5	\$16,660	313	\$20,662	0.464	1.00	0.00

Fig. 3. Simulation result of the hydrokinetic.

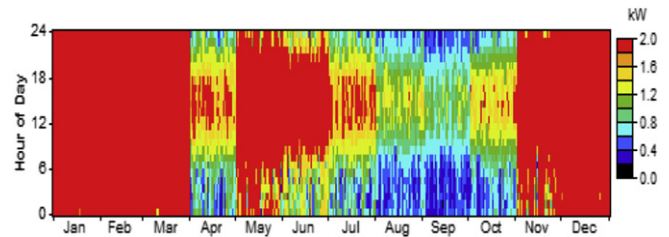


Fig. 4. Hydrokinetic output.

4.3.3. Wind system

The wind speed variations are of great impact on the energy availability produced by the system. Thus, wind turbine rating is usually much higher compared to the average electrical power demand. For our study, we have considered the XLR turbine manufactured by Bergey Windpower and rated at 7.5 kW. The cost of the system is \$26,900, the replacement and maintenance costs are taken as \$15,000 and \$75/year [23]. The lifetime of the wind turbine is taken as 20 year.

4.3.4. Diesel generators

Given that for the rural household (Case 1) the peak power demand is 3.4 kW, the diesel generator cost is taken as \$900. For the BST (Case 2), the price of the diesel generator is taken as 1950 \$ [24]. The operating and maintenance costs are 0.5\$/h and the fuel consumption (0.55 L/kWh). In the South Africa the price of the diesel and lubricant is 1.2 \$/l and 1.30 \$/l respectively.

We have also to take in account the international carbon emission penalty of 2.25 \$/t.

4.3.5. Simulation results and discussion

HOMER simulates system configurations with all of the combinations of components that were specified in the component input. It discards from the results, all non-feasible system configurations, which are those that do not adequately meet the load, given either the available resource or constraints that were specified.

The simulation results will be analyzed and then compared to those acquired by the use of the PV, wind standalone, diesel generator as well as hybrid diesel-battery used to supply the same load. The comparison criteria will be the Initial Capital (IC), the Total Net Present Cost (NPC), the Cost of Energy (COE) as well as the system Capacity Shortage.

4.4. Case study 1: rural household

4.4.1. Hydrokinetic

The architecture and costs of the hydrokinetic option found feasible by Homer are presented on Fig. 3. For the selected site, the optimal combination of 2 hydrokinetic modules, 4 batteries and a 3.5 kW converter has an IC of \$16,660; an NPC of \$20,662, an OC of \$313/yr and a COE of 0.464 \$/kWh.

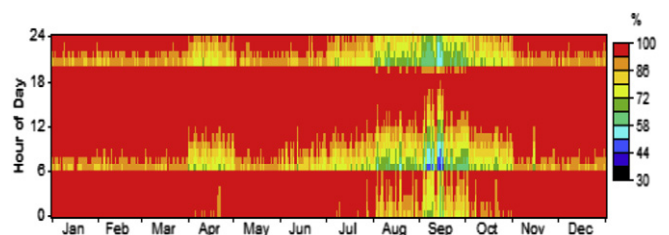


Fig. 5. Battery state of charge.

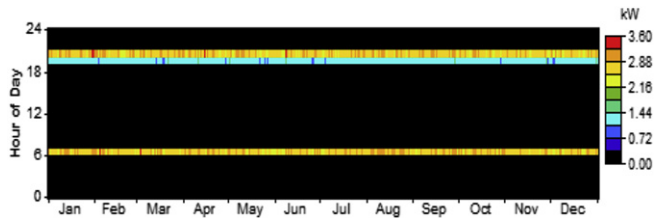


Fig. 6. Inverter output.

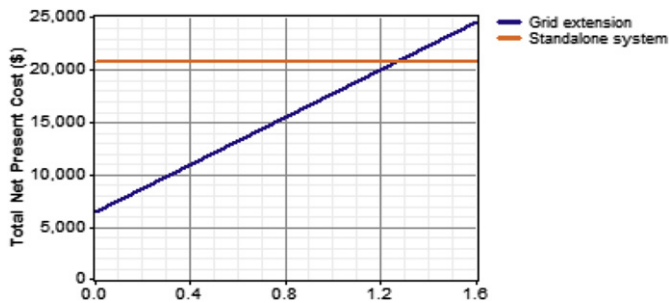


Fig. 7. Grid extension distance (km).

	PV (kW)	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
	5	23	3.5	\$28,245	1,388	\$45,989	1.036	1.00	0.00

Fig. 8. Simulation result of the PV.

Fig. 4 shows the average monthly hydrokinetic output power. During the month of September, due to insufficient water resource the hydrokinetic plan gives an average of 1 kW which is its minimum output.

Figs. 5 and 6 show the converter usage and the battery state of charging. It can easily be seen that from April to November the battery system is charged during off-peak times and used during peak power demand times occurring in the mornings and evenings. Fig. 6 shows that the power from the battery is used only during morning and evening peak times to compensate the hydrokinetic deficit in power supply.

Fig. 7 gives the breakeven grid extension distance at 0.911 km. This means that the total cost of implementing the micro-hydro project for 25 years will be equivalent to the cost of installing a grid extension line of 0.911 km.

4.4.2. Photovoltaic system

From Fig. 8 the optimal size and configuration of the pure PV system are 5 kW PV, 23 batteries and 3.5 kW inverter. It has an IC of \$28,245, an NPC of \$45,989, an OC of \$1388/yr and a COE of 1.036 \$/kWh.

It is noticeable that the cost of 1 kWh produced by the PV is 2.5 times higher compared to the cost of energy produced by the hydrokinetic system.

	XLR	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
	7	357	3.4	\$261,475	11,882	\$413,362	9.311	1.00	0.00

Fig. 9. Simulation result of the wind system.

	Label (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage	Diesel (L)	Label (hrs)
	3.4	\$900	10,821	\$139,234	3.125	0.00	0.00	4,952	8,760

Fig. 10. Simulation result of the diesel generator.

4.4.3. Wind energy system

From Fig. 9 the optimal size and configuration of the pure wind energy system are 7 wind turbine of 7.5 kW each, 357 batteries and 3.5 kW inverter. It has an IC of \$261,475, an NPC of \$413,362, an OC of \$11,882/yr and a COE of 9.311 \$/kWh.

We can easily see that small wind turbines are not particularly efficient and need to be situated in an area of above average wind in order to generate reasonable amounts of power.

4.4.4. Diesel generator

From Fig. 10 the optimal size of the diesel generator is 3.4 kW. It has an IC of \$900, an NPC of \$139,234, an OC of \$10,821/yr and a COE of 3.125 \$/kWh.

The cost of 1 kWh produced by the diesel generator is 7 times higher compared to the cost of energy produced by the hydrokinetic system.

By using the hybrid system rather the diesel generator standalone, 4952 L corresponding to \$5943 can be saved annually. Table 6 shows the different pollutants emitted per year by using the diesel generator which have impacts such as global warming and on the environment in general. For the duration of the project (25 years) the use of the hydrokinetic is a more environmental friendly solution to supply the load compare to the diesel generator.

4.4.5. Case study 1 summary

A summary of the technical and economical results obtained by Homer is displayed on Table 7. From this table we can notice that based on the NPV, COE and the breakeven grid extension distance that the hydrokinetic is the best option to supply the load with electricity.

4.5. Case study 2: BTS load

A similar analysis done with case 1 has also been done with the case 2 (BTS load) under different load and resources conditions.

4.5.1. Hydrokinetic

The architecture and costs of the hydrokinetic option found feasible by Homer are presented on Fig. 3. For the selected site, the optimal combination of 4 hydrokinetic modules, 6 batteries and a 7.5 kW converter has an IC of \$37,050; an NPC of \$53,087, an OC of \$641/yr and a COE of 0.1 \$/kWh.

Fig. 4 shows the average monthly hydrokinetic output power which is close to maximum value all along the year.

Figs. 5 and 6 show the converter usage and the battery state of charging. We can notice that the power from the battery is mainly used in conjunction with the hydrokinetic mainly during the evening peak demand to supply the load.

Table 6
Diesel generator emissions.

Pollutant	Emissions (kg/yr)
Carbon dioxide	13,042
Carbon monoxide	32.2
Unburned hydrocarbons	3.57
Particulate matter	2.43
Sulfur dioxide	26.2
Nitrogen oxides	287

Table 7
Simulation results summary (Case 1).

Costs	HKP	PV	Wind	DG
Capital (\$)	16,660	28,245	261,475	900
Replacement (\$)	3290	13,611	143,869	6456
O&M (\$)	895	8309	31,306	55,991
Fuel (\$)	0	0	0	75,971
Salvage (\$)	–183	–4176	–23,288	–84
Total NPC (\$)	20,662	45,989	413,363	139,234
COE (\$/kWh)	0.464	1.036	9.311	3.125
Grid extension (km)	1.26	3.5	36	11.8

G1	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
4	6	7.5	\$37,050	641	\$53,087	0.100	1.00	0.00

Fig. 11. Simulation result of the hydrokinetic.

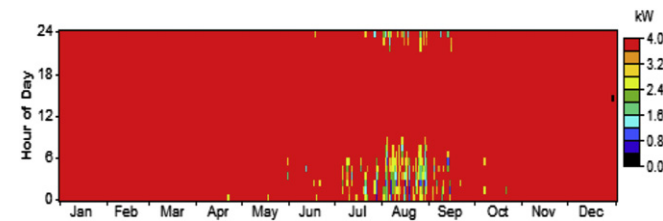


Fig. 12. Hydrokinetic output.

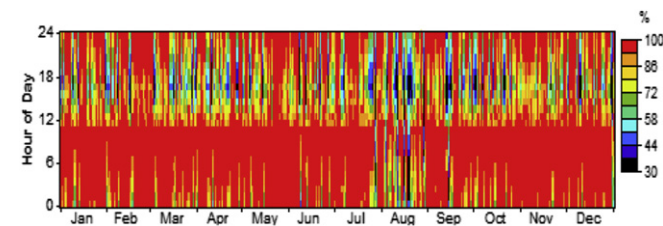


Fig. 13. Battery state of charge.

4.5.2. Photovoltaic system

From Figs. 11–15 the optimal size and configuration of the pure PV system are 30 kW PV, 134 batteries and 7.5 kW inverter. It has an IC of \$152,450, an NPC of \$301,025, an OC of \$5943/yr and a COE of 0.568 \$/kWh.

It is noticeable that the cost of 1 kWh produced by the PV is 5.6 times higher compared to the cost of energy produced by the hydrokinetic system for this specific load and resources.

4.5.3. Wind energy system

From Figs. 16 and 17 the optimal size and configuration of the pure wind energy system are 4 wind turbine of 7.5 kW each, 218

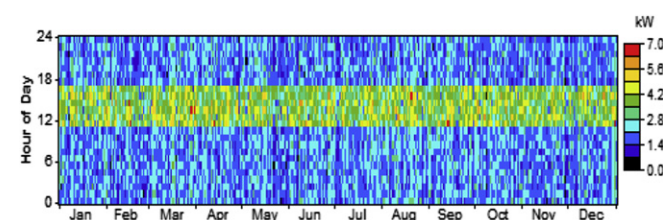


Fig. 14. Inverter output.

PV (kW)	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
30	134	7.5	\$152,450	5,943	\$301,025	0.568	1.00	0.00

Fig. 15. Simulation result of the PV.

XLR	L16P	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage
4	218	7.5	\$151,750	3,051	\$228,025	0.431	1.00	0.00

Fig. 16. Simulation result of the wind system.

Label (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage	Diesel (L)	Label (hrs)
7.1	\$1,950	15,003	\$377,037	0.709	0.00	0.00	10,753	8,759

Fig. 17. Simulation result of the diesel generator.

Table 8
Diesel generator emissions.

Pollutant	Emissions (kg/yr)
Carbon dioxide	28,316
Carbon monoxide	69.9
Unburned hydrocarbons	7.74
Particulate matter	5.27
Sulfur dioxide	56.9
Nitrogen oxides	624

batteries and 7.5 kW inverter. It has an IC of \$151,750, an NPC of \$228,025, an OC of \$3051/yr and a COE of 0.431 \$/kWh.

For this specific area we can see that the wind system is a good option to supply the BTS load compared to the PV system. But its cost still 4.3 times higher compared to 1 kWh produced by the hydrokinetic system.

4.5.4. Diesel generator

The cost of 1 kWh produced by the diesel generator is 7 times higher compared to the cost of energy produced by the hydrokinetic system. By using the hybrid system rather the diesel generator standalone, 10,753 L corresponding to \$13,441 can be saved annually.

Table 8 shows the different pollutants emitted per year by using the diesel generator which have impacts such as global warming and on the environment in general.

4.5.5. Case study 2 summary

As for the case 1, a summary of the technical and economical results obtained by Homer is displayed on Table 9. From this table we can notice that based on the Net Present Cost (NPC), Cost of Energy (COE) and the breakeven grid extension distance that the hydrokinetic is the best option to supply the load with electricity.

Table 9
Simulation results summary (Case 2).

Costs	HKP	PV	Wind	DG
Capital (\$)	37,050	152,450	151,750	1950
Replacement (\$)	11,400	151,200	71,400	27,300
O&M (\$)	9825	90,675	25,725	12,542
Fuel (\$)	0	0	0	336,028
Salvage (\$)	–5188	–93,300	–20,850	–783
Total NPC (\$)	53,087	301,025	228,025	377,037
COE (\$/kWh)	0.100	0.568	0.431	0.709
Grid extension (km)	0.00929	20.7	14.6	27

5. Conclusion

This paper aimed to investigate the possibility of using and developing hydrokinetic power suitable to supply electricity to rural and isolated loads in South Africa where reasonable water resource is available.

The proposed hydrokinetic system is sized to meet the load energy requirement during the worst months. Simulations of the hydrokinetic power have been performed with HOMER software with a rural household and a BTS load as case studies under different demand and energy resources. The results have been compared with those from a diesel generator, wind turbine and standalone PV system while they are supplying the same load. The comparison criteria were the Initial Capital, the Total Net Present Cost, the Cost of Energy as well as the system Capacity Shortage. In summary, hydrokinetic power generation is the best supply option compared to the wind, PV and diesel generator where adequate water resource is available. Apart for being very cost effective, the hydrokinetic system contributes to the reduction of the CO₂ and green house gases in the atmosphere.

The results of this study have led to the following further study recommendations:

- Identify more sites in addition to those already identified for traditional micro-hydropower, and assess potential energy available,
- Develop policies supporting the development and deployment of hydrokinetic power in South Africa.

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